

Saria Al Lahham

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SUMMARY

Senior AI/ML Engineer with 5+ years of experience in Deep Learning, Generative AI, LLMs, and agentic systems. Currently AI Engineering Lead at AlexaTranslations, leading the design and deployment of agent-based LLM systems, RAG pipelines, and MLOps workflows for production translation. Previously AI/ML Research Engineer at Samsung Research America (AI Center, Montreal), working on embedded AI, wireless communications (5G/6G), indoor localization, and human-state estimation.

EDUCATION

Queen's University

Kingston, ON, Canada

Ph.D. in Computer Science; GPA: 4.23/4.30

Thesis: Quantifying Computational Reliability in Extreme Edge Computing: Applications to Distributed Edge AI

Queen's University

Kingston, ON, Canada

M.Sc. in Computer Science; GPA: 4.30/4.30

Thesis: Multi-Orchestrator Mobile Edge Learning: Designing Energy-Efficient Task Allocation and Incentive Schemes

Qatar University

Doha, Qatar

B.Sc. in Computer Engineering; GPA: 3.90/4.00

Graduated with High Order of Excellence

Senior Project: Designing a Smart Home Controller for Smart Home Devices using Hand Gestures

JOB EXPERIENCE

AlexaTranslations

Montreal, QC, Canada

AI Engineering Lead

Jul 2025 – Present

- Architected AI services across the product suite, defining system design and orchestration for production LLM and agentic workloads.
- Developed sovereign AI capabilities addressing PII, sensitive data classification, data residency, and legal constraints around model training, fine-tuning, and inference.
- Translated business problems into scalable AI solutions, partnering with cross-functional teams to align technical strategy with product and operational needs.
- Proposed and drove new AI initiatives, surfacing opportunities to Product and owning the full lifecycle from idea to client-facing product or feature.

AI Engineer

Jul 2024 – Jun 2025

- Designed and deployed a RAG-powered translation system with agentic LLM orchestration, outperforming commercial translation engines on in-house data.
- Designed and implemented automated metric-collection and quality-evaluation pipelines for continuous production monitoring.
- Evaluated recent GenAI and LLM methods, benchmarking against production systems to guide architecture decisions.
- Collaborated with Product and Linguistic Operations teams to deploy features, evaluate system outputs, and curate training data.

Samsung Research America, AI Center

Montreal, QC, Canada

AI/ML Research Engineer

Oct 2022 – Feb 2024

- Developed, deployed, and tested embedded AI algorithms on real hardware and simulations for indoor localization, computer vision, and RAN optimizations.
- Translated mathematical and algorithmic specifications into efficient deployable code.
- Authored scientific papers and patent applications.
- Collaborated with Samsung business units to identify and scope new AI-driven projects.

Queen's University, Telecommunications Research Lab

Graduate Research Fellow

Kingston, ON, Canada

Jan 2021 – Sep 2022

- Modeled and analyzed Federated Learning at the network edge for resource-limited smart devices.
- Developed energy-efficient protocols for Federated Learning.
- Authored and reviewed research articles.

Qatar University, Smart IoT Systems Lab

Research Assistant

Doha, Qatar

May 2020 – Sep 2021

- Designed and implemented AI-based protocols for Ultra Reliable Low Latency Communication (URLLC) in smart health systems.
- Surveyed and integrated recent algorithms into protocol designs.
- Authored and reviewed research articles.

SKILLS & EXPERTISE

Software Development

- **Programming Languages:** Python, MATLAB, C/C++, Java, and Bash scripting.
- **Machine Learning:** SciPy, NumPy, SciKit-Learn, Pandas, XGBoost, Matplotlib and Seaborn.
- **Deep Learning:** PyTorch, Tensorflow/Keras, JAX, OpenCV, and Vision Transformers.
- **Generative AI:** SGLang, vLLM, RAG, vector databases, HuggingFace, Unsloth, OpenAI/Gemini APIs.
- **MLOps:** LightningAI, Weights & Biases, Docker, AWS Cloud, OpenSearch and Azure Cloud.
- **Simulation & Digital Twins:** Nvidia Sionna, Blender

Artificial Intelligence

- Data Science, Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Multimodal Foundation Models, RAG, Knowledge Graphs, Agentic AI, Reinforcement Learning, and Multi-Agent Systems.

Telecommunications and Computer Networks

- Digital Signal Processing, Edge Networks, IoT, TCP/IP, Wireless and Cellular Networks (5G/6G), 3GPP/IEEE/ETSI Standards, and Digital Twins.

PATENTS

US 2024/0422642 A1 — Probabilistic Mobility Load Balancing in Multi-Band Mobile Communication Network

Dec 19, 2024

U.S. Patent Application No. 18/593,262 (Published). Applicant: Samsung Electronics Co., Ltd

- **Inventors:** MHD Saria Al Lahham; Di Wu; Abu Zafar Ekram Hossain; Xue Liu; Gregory L. Dudek

LIST OF PUBLICATIONS

- [1] Alaa Awad Abdellatif, **Allahham, Mhd Saria**, Noor Khial, Amr Mohamed, Aiman Erbad, and Khaled Shaban. Reliable federated learning for age sensitive mobile edge computing systems. In *ICC 2023-IEEE International Conference on Communications*, pages 1622–1627. IEEE, 2023.
- [2] Alaa Awad Abdellatif, **Allahham, Mhd Saria**, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. Onsra: An optimal network selection and resource allocation framework in multi-rat systems. In *ICC 2021-IEEE International Conference on Communications*, pages 1–6. IEEE, 2021.
- [3] Sherif B. Azmy, **Allahham, Mhd Saria**, Nizar Zorba, and Hossam S. Hassanein. Quantifying the impact of incentives on service availability at the extreme edge. In *GLOBECOM 2024 - 2024 IEEE Global Communications Conference*, pages 4162–4167, 2024.
- [4] Emna Baccour, **Allahham, Mhd Saria**, Aiman Erbad, Amr Mohamed, Ahmed Refaey Hussein, and Mounir Hamdi. Zero touch realization of pervasive artificial intelligence as a service in 6g networks. *IEEE Communications Magazine*, 61(2):110–116, 2023.
- [5] Yahuza Bello, Alaa Awad Abdellatif, **Allahham, Mhd Saria**, Ahmed Refaey Hussein, Aiman Erbad, Amr Mohamed, and Mohsen Guizani. B5g: Predictive container auto-scaling for cellular evolved packet core. *IEEE Access*, 9:158204–158214, 2021.

- [6] Heba DM Dawoud, **Allahham, Mhd Saria**, Alaa Awad Abdellatif, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. Patient-driven network selection in multi-rat health systems using deep reinforcement learning. In *2021 IEEE Global Communications Conference (GLOBECOM)*, pages 1–6. IEEE, 2021.
- [7] Noor Khial, **Allahham, Mhd Saria**, Naram Mhaisen, Loay Ismail, Mohamed Mabrok, and Amr Mohamed. Drone-rl: Dynamic reinforcement learning for online navigation of uavs in evolving environments. *Knowledge-Based Systems*, page 115147, 2025.
- [8] Naram Mhaisen, **Allahham, Mhd Saria**, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. On designing smart agents for service provisioning in blockchain-powered systems. *IEEE Transactions on Network Science and Engineering*, 9(2):401–415, 2021.
- [9] Hassan Saadat, **Allahham, Mhd Saria**, Alaa Awad Abdellatif, Aiman Erbad, and Amr Mohamed. RL-assisted energy-aware user-edge association for iot-based hierarchical federated learning. In *2022 International Wireless Communications and Mobile Computing (IWCMC)*, pages 548–553. IEEE, 2022.
- [10] **Al Laham, Saria**, Bobak H Baghi, Pierre-Yves Lajoie, Amal Feriani, Sachini Herath, Steve Liu, and Gregory Dudek. Device-free human state estimation using uwb multi-static radios. *arXiv preprint arXiv:2401.05410*, 2023.
- [11] **Allahham, Mhd Saria**, Alaa Awad Abdellatif, Naram Mhaisen, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. Multi-agent reinforcement learning for network selection and resource allocation in heterogeneous multi-rat networks. *IEEE Transactions on Cognitive Communications and Networking*, 8(2):1287–1300, 2022.
- [12] **Allahham, Mhd Saria**, Alaa Awad Abdellatif, Amr Mohamed, Aiman Erbad, Elias Yaacoub, and Mohsen Guizani. I-see: Intelligent, secure, and energy-efficient techniques for medical data transmission using deep reinforcement learning. *IEEE Internet of Things Journal*, 8(8):6454–6468, 2020.
- [13] **Allahham, Mhd Saria**, Mohammad F Al-Sa’d, Abdulla Al-Ali, Amr Mohamed, Tamer Khattab, and Aiman Erbad. Dronerf dataset: A dataset of drones for rf-based detection, classification and identification. *Data in brief*, 26:104313, 2019.
- [14] **Allahham, Mhd Saria**, Salimur Choudhury, and Hossam S. Hassanein. Reliable federated learning with auction-based incentives at the extreme edge. In *GLOBECOM 2024 - 2024 IEEE Global Communications Conference*, pages 3134–3139, 2024.
- [15] **Allahham, Mhd Saria** and Hossam S. Hassanein. Quantifying computational reliability of task assignment in extreme edge computing. *IEEE Transactions on Mobile Computing*, pages 1–15, 2026.
- [16] **Allahham, Mhd Saria**, Tamer Khattab, and Amr Mohamed. Deep learning for rf-based drone detection and identification: A multi-channel 1-d convolutional neural networks approach. In *2020 IEEE International Conference on Informatics, IoT, and Enabling Technologies (ICIOT)*, pages 112–117. IEEE, 2020.
- [17] **Allahham, Mhd Saria**, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. Motivating learners in multi-orchestrator mobile edge learning: A stackelberg game approach. *IEEE Canadian Journal of Electrical and Computer Engineering*, 46(1):69–76, 2022.
- [18] **Allahham, Mhd Saria**, Amr Mohamed, Aiman Erbad, and Hossam Hassanein. On the modeling of reliability in extreme edge computing systems. In *2022 5th International Conference on Communications, Signal Processing, and their Applications (ICCSPA)*, pages 1–6. IEEE, 2022.
- [19] **Allahham, Mhd Saria**, Amr Mohamed, and Hossam Hassanein. Incentive-based resource allocation for mobile edge learning. In *2022 IEEE 47th Conference on Local Computer Networks (LCN)*, pages 157–164. IEEE, 2022.
- [20] **Allahham, Mhd Saria**, Sameh Sorour, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. Energy-efficient multi-orchestrator mobile edge learning. *arXiv preprint arXiv:2109.00757*, 2021.

- [21] **Lahham, Saria Al**, Di Wu, Ekram Hossain, Xue Liu, and Gregory Dudek. Probabilistic mobility load balancing for multi-band 5g and beyond networks. In *2024 IEEE International Conference on Communications Workshops (ICC Workshops)*, pages 1673–1678, 2024.